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Description - Procedure for replacing the entire RF amplifier, which is located in the RF/PEN Cabinet.

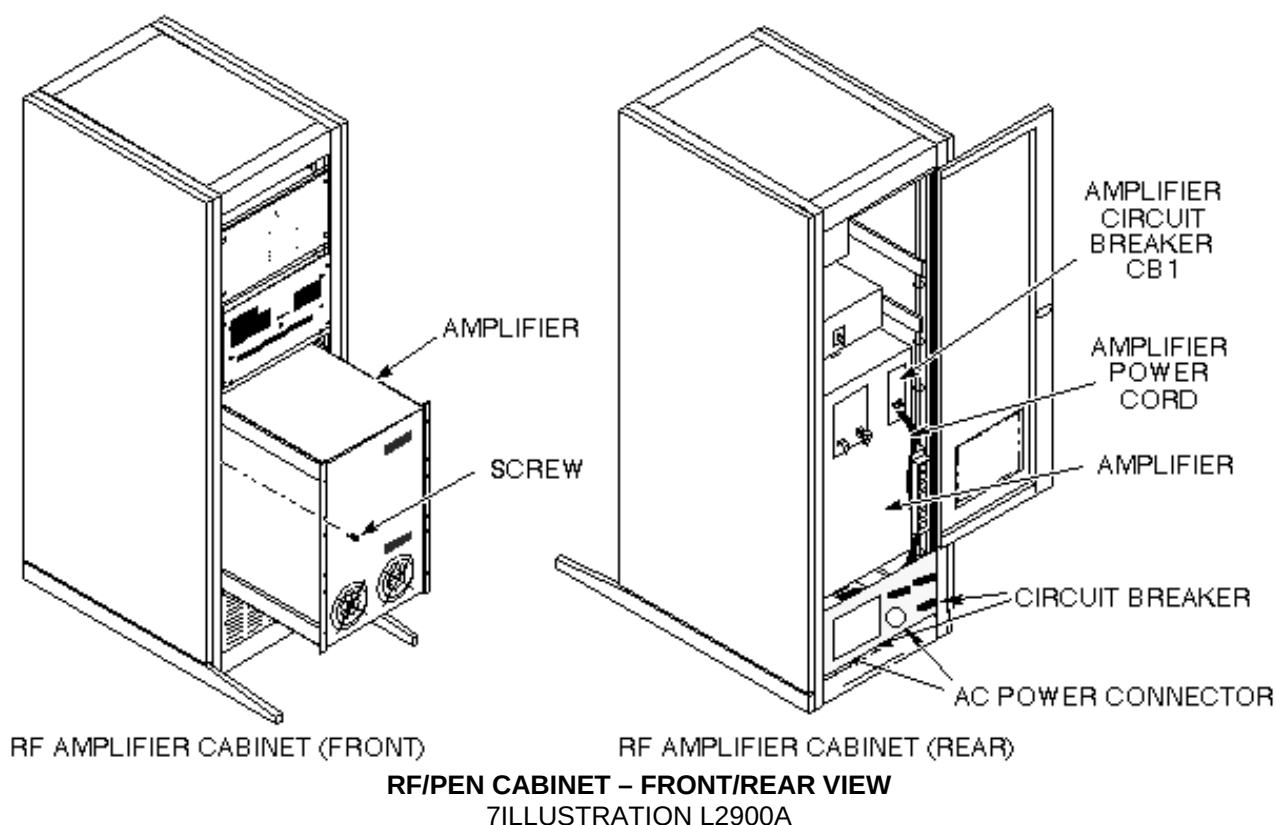
For individual RF amplifier module replacement procedures, refer to:

1.5T: Procedure for 20 kW RF Amplifier Replacement

1.0T: Procedure for 8 kW RF Amplifier Replacement

1- RF AMPLIFIER PREPARATION

1. Switch off RF/PEN Cabinet circuit breaker CB6, at rear of cabinet. See Illustration L2900A.



2. Remove ac power cable at rear of cabinet. Lock out and tag out the ac power connector.
See Illustration L2900A above.
(Refer to *Procedure For Safety: Section 6*)

WARNING!

**POSSIBLE PERSONAL INJURY! POWER TO THE ERBTEC AMPLIFIER
MUST BE TURNED OFF, LOCKED OUT, AND TAGGED OUT BEFORE YOU
PROCEED.**

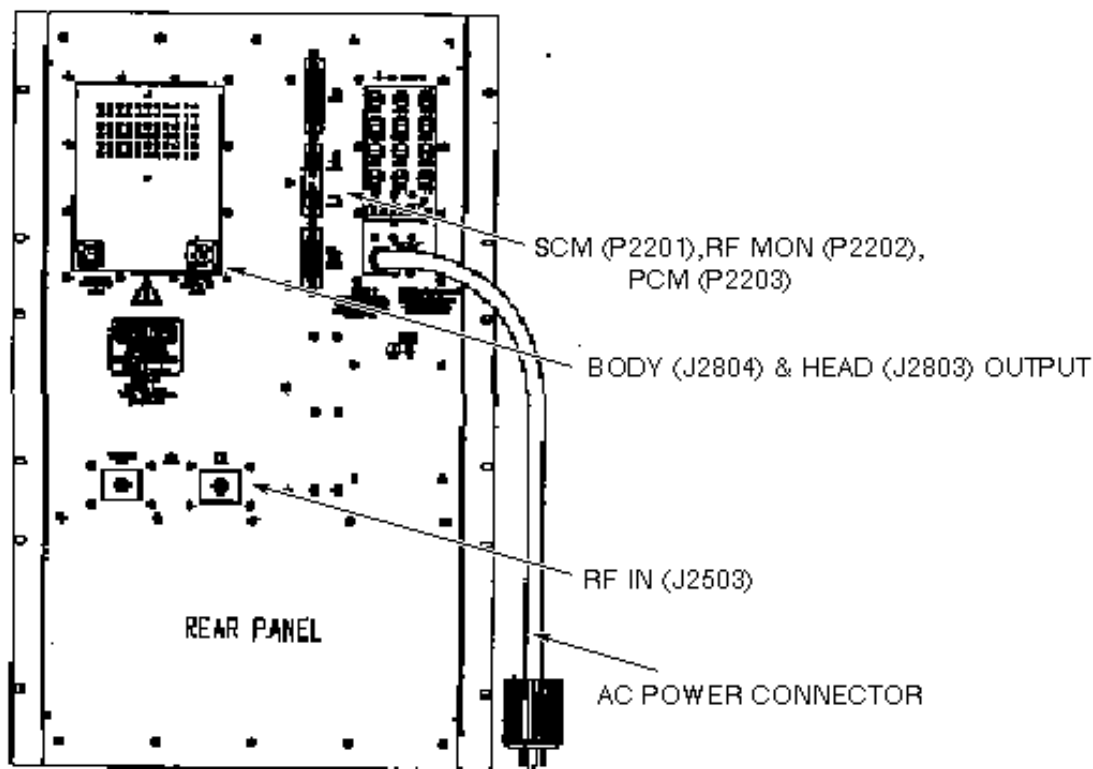
3. Remove front panel of the RF/PEN Cabinet.

- 4 Verify that power to the Erbttec RF Amplifier is off by checking the fans and LEDs.
5. Remove the twelve screws from the front of the amplifier (see Illustration L2900A above).

WARNING!

POSSIBLE PERSONAL INJURY OR EQUIPMENT DAMAGE! THE CABINET MUST HAVE TWO ANTI-TIP LEGS ATTACHED BEFORE YOU ATTEMPT TO REMOVE THE AMPLIFIER.

6. Slide the Erbttec RF Amplifier forward until the side panels are fully exposed. See Illustration L2900A above.
7. Open the RF/PEN Cabinet rear cabinet door.
8. Disconnect ac power cable and other connectors from rear panel of amplifier (see Illustration L2901A).



RF AMPLIFIER – REAR VIEW
ILLUSTRATION L2901A

2- LIFT TOOL ASSEMBLY & INSTALLATION

WARNING!

POSSIBLE PERSONAL INJURY OR EQUIPMENT DAMAGE! THE ERBTEC RF AMPLIFIER WEIGHS MORE THAN 275 LBS. TWO PEOPLE ARE REQUIRED TO REMOVE AMPLIFIER ASSEMBLY FROM CABINET IF LIFT TOOL IS NOT USED.

WARNING!

POSSIBLE PERSONAL INJURY. HEAVY OBJECT - THE HOIST KIT WEIGHS 73 LBS/33 KG. ASSISTANCE TO LIFT THIS PRODUCT [MECHANICAL (DOLLIES) OR ADDITIONAL PERSONNEL] IS REQUIRED.

Note

RF Lift Tool - The RF Lift Tool permits replacement of an Erbtec RF Amplifier by one person. This is the preferred method of replacement to avoid personal injury.

Required Materials

There are two RF Lift Tool Kits:

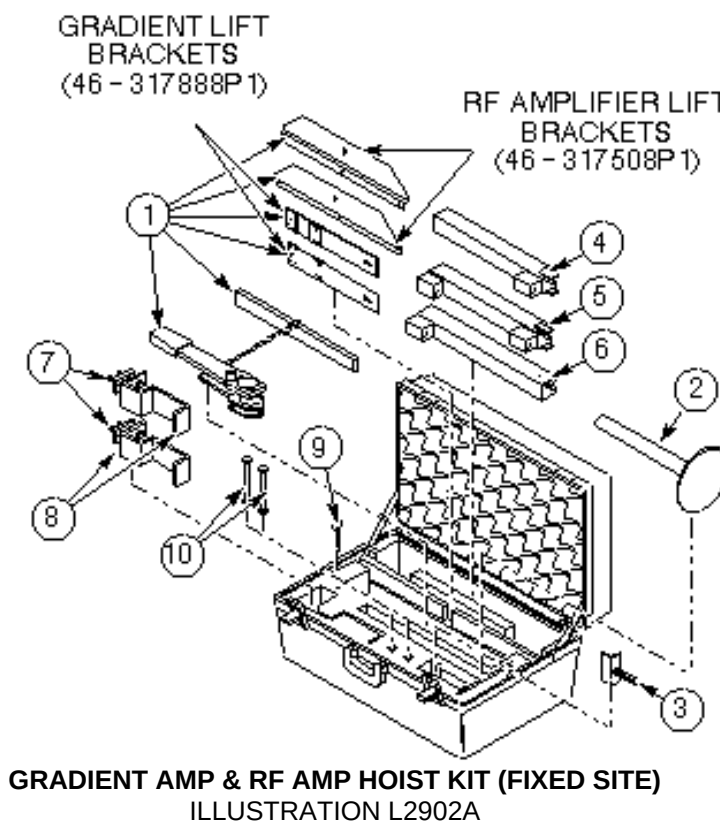
- Fixed site 46-317266G1, see Table 1 and Illustration L2902A.

TABLE 1
FIXED SITE RF LIFT TOOL KIT 46-317266G1

Item	Part Number	Description	Quantity
1.	46-195700G4	Hoist Assembly	1
2.	46-317263P1	Support Jack	1
3.	46-317289G1	Jack Head for Support Jack	2
4.	46-317258P1	Rail, Front Section	1
5.	46-317259P1	Rail, Center Section	1
6.	46-317451P1	Rail, Right Section	1
7.	46-317262P1	Plate (for Bracket)	2
8.	46-317280G1	Bracket	2
9.	46-221889P8	Lock Pin	1
10.	46-208561P139	Cap Screw, 1/2"-13x4.75"	2
11.	46-208932P2	Cap Screw, 1/4"-20x1.00"	4
12.	46-208948P3	Hex Nut, 1/2"	2
13.	46-317738P2	6" Support Jack Extension for Support Jack	1

Note

Item 13 is not shown in the Illustration.



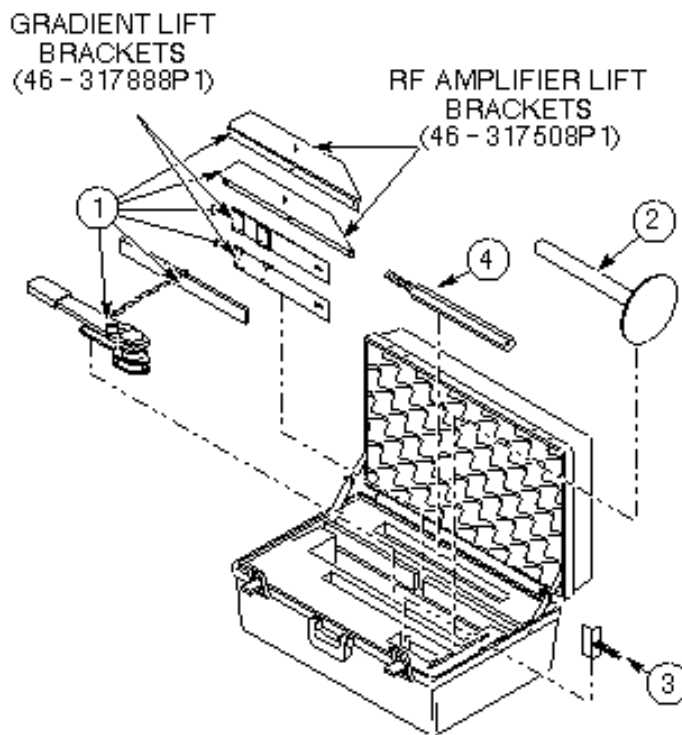
- Mobile site 46-317266G2, see Table 2 and Illustration L2903A.

TABLE 2
MOBILE SITE RF LIFT TOOL KIT 46-317266G2

Item	Part Number	Description	Quantity
1.	46-195700G4	Hoist Assembly	1
2.	46-317263P1	Support Jack	1
3.	46-317289G1	Jack Head for Support Jack	2
4.	46-317738P1	Support Jack Extension for Support Jack	1
5.	46-317738P2	6" Support Jack Extension for Support Jack	1

Note

Item 5 is not shown in the Illustration.



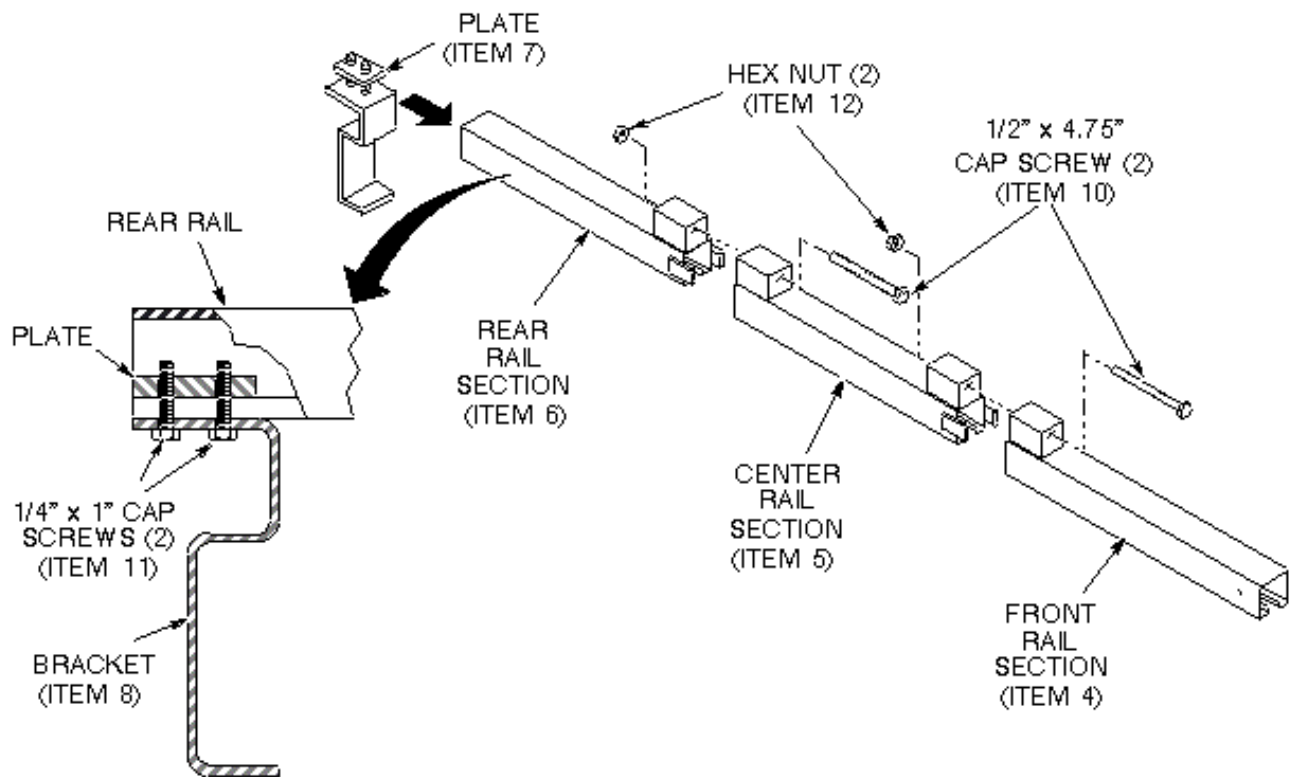
GRADIENT AMP & RF HOIST KIT (MOBILE SITE)
ILLUSTRATION L2903A

Note

Steps 1-8 are for fixed sites only - Steps 1-8 are for fixed sites only. Mobile sites begin with step 9.

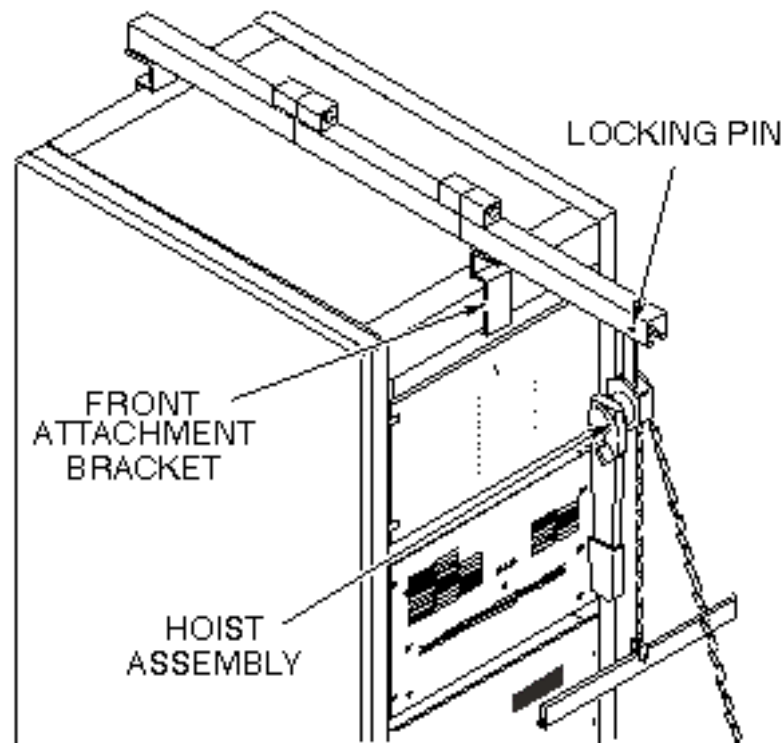
Procedure

1. Assemble three piece rail (Items 4, 5 and 6) with cap screws (Item 10) and nuts (Item 12). See Illustration L2904A.



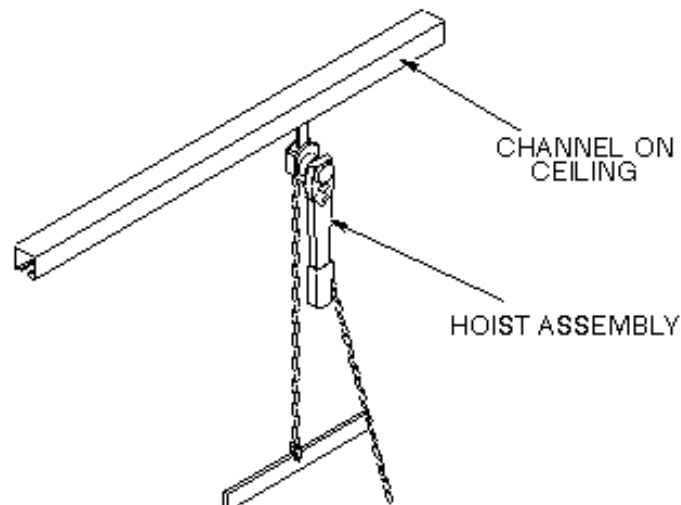
LIFT TOOL ASSEMBLY (FIXED SITE)
ILLUSTRATION L2904A

2. Assemble bracket (Item 8) to plate (Item 7) with two cap screws (Item 11). See Illustration L2904A above.
3. Insert plate (Item 7) into rear rail section (Item 6) until flush and tighten cap screws (Item 11).
4. Remove front cover from RF/PEN Cabinet.
5. Remove front cover upper attachment bracket (if installed).
6. Place three-piece rail assembly on top of cabinet with attached bracket to rear of cabinet. See Illustration L2905A.



LIFT TOOL ATTACHMENT (FIXED SITE)
ILLUSTRATION L2905A

7. Assemble bracket (Item 8) to plate (Item 7) with two cap screws (Item 11).
8. Insert late (Item 7) into front rail section (Item 4), push bracket against cabinet front and tighten cap screws (Item 11). See Illustration L2905A above.
9. Insert hoist assembly pivot bearings, with handle at left, into front rail section for fixed site, see Illustration L2905A above or ceiling channel for mobile site, see Illustration L2906A.



LIFT TOOL ATTACHMENT (MOBILE SITE)
ILLUSTRATION L2906A

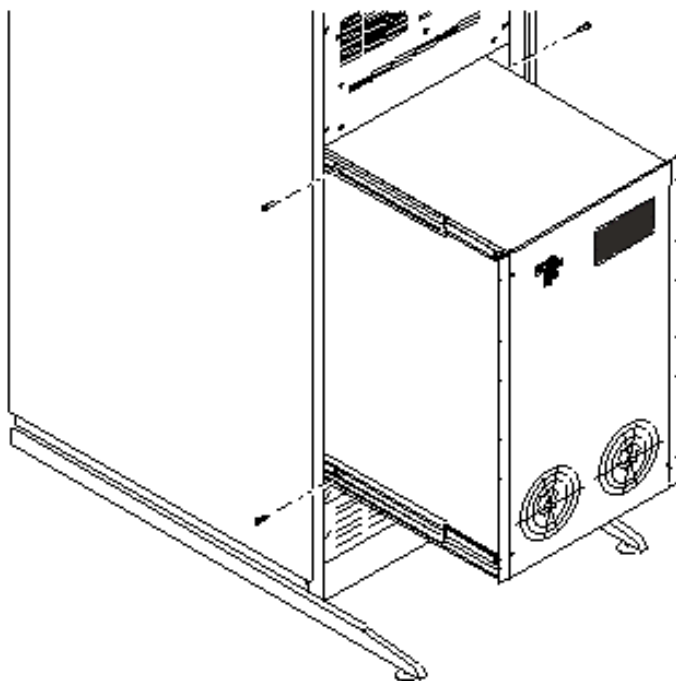
10. Insert lock pin (Item 9) through holes in front rail section.

3- REMOVING FAILED RF AMPLIFIER



Possible personal injury or equipment damage. Remove only four rear drawer slide screws at this time, until support stand is in place. After the stand is in place, the other screws may be removed.

1. Pull out RF drawer approximately eighteen inches. See Illustration L2907A.



REMOVAL OF SCREWS THRU ACCESS HOLES
ILLUSTRATION L2907A

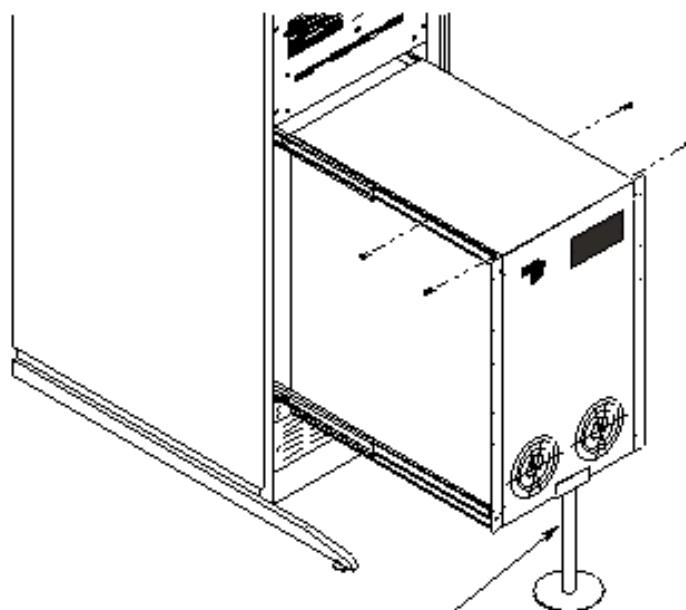
2. Pull top drawer slide on RF amp forward until the third from front attachment screw head is exposed through the access hole in the slide.
3. Remove screw.
4. Repeat steps 2 and 3 for other top and bottom drawer slide screws.



POSSIBLE PERSONAL INJURY OR EQUIPMENT DAMAGE! THE CABINET MUST HAVE TWO ANTI-TIP LEGS ATTACHED BEFORE YOU ATTEMPT TO REMOVE THE AMPLIFIER.

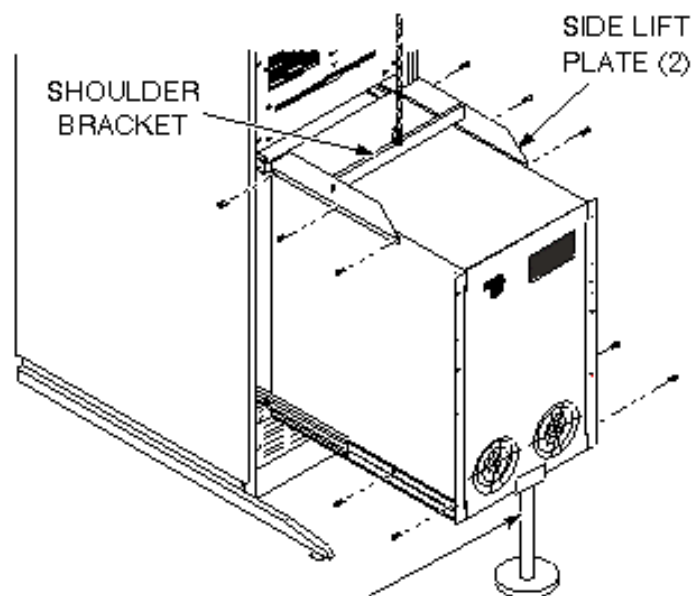
5. Pull out RF drawer to lock detent (full out).

6. Place support stand under front edge of drawer and adjust to put slight upward force on drawer. Include 6-inch support jack extension, 46-317738P2 (see Illustration L2908A).



SUPPORT JACK
SUPPORT TOOL USE
ILLUSTRATION L2908A

7. Remove remaining four outer screws from two top drawer channels.
8. Attach side lift plate to right rear of RF amp with three original RF amp screws. See Illustration L2909A.



SUPPORT JACK
SIDE BRACKET ATTACHMENT
ILLUSTRATION L2909A

9. Insert shoulder bracket screw into right side lift bracket hole.
10. Attach left side lift plate to the rear of RF amp while keeping the shoulder bracket screw in the lift plate hole.
11. Pull upward on shoulder bracket to seat two shoulder screws.
12. Apply slight tension to support chain with chain hoist.
13. Remove remaining four screws from bottom drawer channels.
14. The RF amplifier should be hanging freely from the chain at this time.

4- INSTALLING REPLACEMENT RF AMPLIFIER

1. Install Lift Tool Assembly on RF/PEN Cabinet. Refer to Section 2-.
2. Attach side lift plate to right side of drawer with three original RF amp screws at the rear of the RF amp (see Illustration L2909A in Section 3).
3. Insert shoulder bracket screw into right side lift bracket hole.
4. Attach left side lift plate to RF amp with three original RF amp screws at the rear of the RF amp while keeping the shoulder bracket screw in lift plate hole.
5. Seat two shoulder screws.
6. Pull upward on shoulder bracket to level bottom drawer channels.
7. Place support jack under front edge of drawer and adjust for alignment of bottom drawer channels.
8. Install four screws into bottom drawer channels.



POSSIBLE PERSONAL INJURY OR EQUIPMENT DAMAGE! THE CABINET MUST HAVE TWO ANTI-TIP LEGS ATTACHED BEFORE YOU ATTEMPT TO REMOVE THE AMPLIFIER.

9. Remove side lift plates and Lift Tool Assembly from RF/PEN Cabinet.
10. Install screws into two top drawer channels. See Illustration L2908A in Section 3.
11. Push in RF drawer to approximately eighteen inches (see Illustration L2907A in Section 3).

12. Pull any drawer slide forward until the third from front attachment screw head is exposed through the access hole in the slide.
13. Install screw.
14. Repeat steps 2 and 3 for other three drawer slides.
15. Install cabling in rear of RF amplifier. See Illustration L2901A in Section 1.
16. Continue with Section 6- Functional Checks Required.

5- RF AMPLIFIER POWER DOWN PREPARATION

1. Command the amp to the Off state (Use *a* or *b* method):

Note

Commanding The Amp To The Off State - Commanding the amp to the Off state, causes the tube(s) to be forward biased, thus dissipating the stored power in the power supply. This process takes 2 to 3 seconds.

- a. With the Serial link cable connected to the RF amplifier, type (o0).
- b. Cycle power to the RF System Controller module (MR1A15). This creates an 11 fault: Do not download TPS.

Note

Clearing An 11 Fault - To clear an 11 fault, you must command the amp to off (o0), even if it indicates OFF. Follow the o0 command with an o1 Standby command. The amp should now recycle in about 30 seconds. The 11 fault must be cleared quickly, or you will have to wait 360 seconds every time you try to recover from one.

2. Wait the 2–3 seconds, then turn the circuit breaker for the Erbttec amp off. This is a second protection.
3. Remove the ac cord for the lock out/tag out procedure. This is a third protection.



USE EXTREME CAUTION IN THE NEXT STEP AS THERE STILL MAY BE HAZARDOUS VOLTAGES PRESENT!!!

4. Remove the respective cover(s). Measure any residual HV potential within the amp, i.e., the HV resistor on the power supply; the case of the PA and/or IPA (if applicable) tube(s).

WARNING!

POSSIBLE PERSONAL INJURY! IF THE MEASURED POTENTIAL IS MORE THAN 20 VOLTS, USE EXTREME CAUTION. HAZARDOUS VOLTAGES MAY BE PRESENT.

Note

DO NOT SHORT The Power Supply To Ground - If this is the case, DO NOT SHORT the power supply to ground. Wait until the voltage drops below 15–20 volts! If there was no problem with the supply before, there will be one if you short it out with too much power in the supply! Protective diodes D1 and/or D12 will be blown.

5. If the measured potential is approximately 15 to 20 volts, short out the respective 3kV, and 6kV supplies. Use a 1/8" x 6" flat blade screwdriver (Xcelite #R186, or Ace #24835) to short the HV standoffs to the chassis.

Note

Prevent Any Build-Up Of Power In The Caps - This is to prevent any build-up of power in the caps, which could be hazardous. If you think the supply is safe, and all of a sudden you get a small zap, you would probably remove your hand quickly, risking injury, or scraping your hands, etc.

Note

If you do not complete the following step, this is where ***you most often expose yourselves to danger.***

6. Immediately after shorting these supplies to ground, **attach a grounding wire to the amp frame first, then attach the opposite end to the power supply HV standoffs.** Two grounding cables are required, one each for the 3 & 6Kv supply.
7. The amp is now safe. Your only other caution would be to remember to use proper ESD practices while working inside the amp.

6- FUNCTIONAL CHECKS REQUIRED

1. For tube-type RF amplifier: Perform Procedure Auto Fine Tune (AFT)
2. Return to Section 5- RF Amplifier Power Down Preparation.
3. **For all RF amplifiers:** Perform one successful head and body scan to verify properly operating system.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	July 21, 1998	J. Saperstein	Initial conversion from Toolbook to Word.
1	May 21, 1999	SM Atladottir	Updated Procedure References for New GUI
2	10/17/2007	K.Keshena	PSR 13105722, requires the warning statement for the Hoist Kit.